

Master/
Orchestrator

SU#

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IJCSA: Solve Algebraic Loop at time t

Initialize U^0 from current inputs

loop

[until convergence]

1. Evaluate subsystems in parallel

par

[Parallel Execution]

evaluate_outputs($U1^m, t$)

$Y1^{m+1} = S1(U1^m)$

Compute with $dt=0$
(frozen state)

$Y1^{m+1}$

evaluate_outputs($U2^m, t$)

$Y2^{m+1} = S2(U2^m)$

Compute with $dt=0$

$Y2^{m+1}$

2. Compute residual

$r^m = [U1^m - Y2^{m+1}]$
 $[U2^m - Y1^{m+1}]$

3. Check convergence

alt

$[||r^m|| < \text{tolerance}]$

✓ Converged

Exit iteration

4. Compute interface Jacobian

$J^m = dR/dU$ via finite differences

For each interface input:
perturb U_j , evaluate outputs,
compute $J[:, j] = (R_{\text{pert}} - R) / \text{eps}$

5. Solve for correction

$J^m \cdot \Delta U^m = -r^m$

6. Apply correction

$U^{m+1} = U^m + \Delta U^m$

7. Commit converged solution

set_inputs($U1^*, t$)

inputs committed

set_inputs($U2^*, t$)

inputs committed

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